Homework 5: Dual Simplex

Problem 1. Consider the linear program.

$$\begin{array}{lrcl} \min & x_1 + x_2 & & \\ \text{s.t.} & x_1 + 2x_2 & \geq & 2 \\ & x_1 & \geq & 1 \\ & x_1, \quad x_2 & \geq & 0. \end{array}$$

a) Write the dual of the linear program.

b) Solve the linear program using the dual simplex method, starting from the initial basic infeasible solution $(x_1, x_2) = (0, 0)$.

Solution to Problem 1.

a) The dual of the linear program is

$$\begin{array}{lrcl} \max & 2\pi_1 + \pi_2 & & \\ \text{s.t.} & \pi_1 + \pi_2 & \leq & 1 \\ & 2\pi_1 & \leq & 1 \\ & \pi_1, \quad \pi_2 & \geq & 0. \end{array}$$

b) Solve the linear program using the dual simplex method.

$$\begin{array}{lrcl} \min & z = \quad x_1 + x_2 & & \\ \text{s.t.} & -x_1 - 2x_2 + s_1 & = & -2 \\ & -x_1 \qquad\quad + s_2 & = & -1 \\ & x_1, \quad x_2, \quad s_1, \quad s_2 & \geq & 0. \end{array}$$

Step 0. (Initialization) Find a basis matrix B for which the corresponding basic solution is dual feasible.
Start with $(x_1, x_2) = (0, 0)$, the basis matrix is

$$B = I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

The initial simplex tableau is as follows.

Basis	x_1	x_2	s_1	s_2	RHS
Row 0	1	1	0	0	0
s_1	-1	-2	1	0	-2
s_2	-1	0	0	1	-1

Note that the zeroth row in the tableau uses $(1, 1, 0, 0)$ instead of $-(1, 1, 0, 0)$ as this is a minimization problem. Correspondingly, the value in the RHS column of the zeroth row represents the negative of the objective function value (i.e., $-z$).

Multiply B^{-1}, obtain the table as following.

Basis	x_1	x_2	s_1	s_2	RHS
Row 0	1	1	0	0	0
s_1	-1	-2	1	0	-2
s_2	-1	0	0	1	-1

$\bar{c}^T = (-1, -1, 0, 0) \leq \mathbf{0}$, the corresponding basic solution is dual feasible.

Iteration 1.

Step 1. Optimality/primal feasibility check.

The primal solution is $x_B = B^{-1}b = \begin{bmatrix} s_1 \\ s_2 \end{bmatrix} = \begin{bmatrix} -2 \\ -1 \end{bmatrix}$, $x_B \leq 0$. Select $l = 1$ with $x_{B(1)} < 0$.

Step 2. Minimum ratio test for dual feasibility.

a) Consider the 1st row of the tableau (the pivot row) $\left[\begin{array}{cccc|c} v_1 & v_2 & v_3 & v_4 & x_{B(l)} \end{array}\right] = \left[\begin{array}{cccc|c} -1 & -2 & 1 & 0 & -2 \end{array}\right]$, there exists $v_1, v_2 < 0$.

b) Compute the minimum ratio $\theta^* = \min_{\{i|v_i<0\}} \frac{\bar{c}_i}{v_i} = \min\{\frac{-1}{-1}, \frac{-1}{-2}\} = \frac{1}{2}$.

c) The x_2 column achieves the minimum ratio, x_2 enters the basis and s_1 exits the basis.

Step 3. Update the basis.

Basis	x_1	x_2	s_1	s_2	RHS
Row 0	1/2	0	1/2	0	-1
x_2	1/2	1	-1/2	0	1
s_2	-1	0	0	1	-1

Iteration 2.

Step 1. Optimality/primal feasibility check.

$x_B = \begin{bmatrix} x_2 \\ s_2 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$, select the $l = 2$ with $x_{B(2)} < 0$.

Step 2. Minimum ratio test for dual feasibility.

a) Consider the 2nd row of the tableau (the pivot row) $\left[\begin{array}{cccc|c} v_1 & v_2 & v_3 & v_4 & x_{B(l)} \end{array}\right] = \left[\begin{array}{cccc|c} -1 & 0 & 0 & 1 & -1 \end{array}\right]$, there exists $v_1 < 0$.

b) Compute the minimum ratio $\theta^* = \min_{\{i|v_i<0\}} \frac{\bar{c}_i}{v_i} = \min\{\frac{-1/2}{-1}\} = 1/2$.

c) The x_1 column achieves the minimum ratio, x_1 enters the basis and s_2 exits the basis.

Step 3. Update the basis.

Basis	x_1	x_2	s_1	s_2	RHS
Row 0	0	0	1/2	1/2	-3/2
x_2	0	1	-1/2	1/2	1/2
x_1	1	0	0	-1	1

Iteration 3.

Step 1. Optimality/primal feasibility check.

$x_B = \begin{bmatrix} x_2 \\ x_1 \end{bmatrix} = \begin{bmatrix} 1/2 \\ 1 \end{bmatrix} \geq 0$, stop. The current solution is primal feasible and thus optimal.

We have

$$x_B^\top = \begin{bmatrix} x_1 & x_2 \end{bmatrix} = \begin{bmatrix} 1 & 1/2 \end{bmatrix}$$
$$x_N^\top = \begin{bmatrix} s_1 & s_2 \end{bmatrix} = \begin{bmatrix} 0 & 0 \end{bmatrix}$$

The optimal solution is $\mathbf{x}^* = (x_1, x_2) = (1, 1/2)$.
The optimal objective value is $z^* = x_1 + x_2 = 1 + 1/2 = 3/2$.